

R. DIVIKA RATHI

Computer Science Student | BINUS University | Jakarta, Indonesia | Email: rdivika.rathi@binus.ac.id

PROFILE

Computer Science student at BINUS University with interests in data analytics, databases, data engineering, and digital products. Experienced in working with SQL, Python, ETL workflows, data visualization, and machine learning through academic and team projects. Interested in applying technical and analytical skills to solve business problems and communicate data-driven insights.

EDUCATION

BINUS University — Bachelor of Computer Science | 2024-Present | GPA: 3.33/4.00

Relevant Coursework: Database Systems, Data Analytics, Data Engineering, Data Visualization, Big Data Processing, Software Engineering

EXPERIENCE

HIMTI BINUS University — Secretary | 2024-Present

- Coordinated documentation, correspondence, schedules, and administrative workflows across organizational activities.
- Collaborated with multiple teams to organize event requirements and maintain clear communication from planning through execution.

BINUS Freshmen Program — Freshmen Partner | Sep 2025-Sep 2026

- Supported freshmen throughout their transition into university life by providing guidance, information, and assistance.
- Collaborated with fellow partners to coordinate communication and support a positive onboarding experience.

BINUS Freshmen Program — Freshmen Leader | Jul 2025-Sep 2025

- Guided incoming freshmen during the university transition period and supported program activities.
- Coordinated with student leaders to communicate information and assist freshmen throughout the program.

DATA & TECHNICAL PROJECTS

Linear Regression & Greedy Join Order Optimization — SQL, MySQL, Python, Scikit-learn

Built a query-cost prediction and join-order optimization framework using 720 MySQL query executions and 17 structural features. Trained Linear Regression on log-transformed execution times and used a Greedy Algorithm to select predicted lower-cost join orders. Achieved R^2 87.45%, RMSE 0.3044 ms, and MAE 0.2206 ms; evaluated 128 join-order scenarios.

Online Retail Data Engineering & Business Intelligence — Python, Pentaho PDI, MySQL, Power BI

Built an end-to-end ETL pipeline for 541,909 Online Retail transactions, including preprocessing, duplicate and missing-data handling, dimensional modeling, and data warehouse loading. Designed a star schema with product, customer, time, and sales fact tables and developed Power BI dashboards for sales, customer behavior, and product performance analysis.

Isyaratku — Django, FastAPI, SQLite, AI, UX/UI

Developed a sign language learning platform combining interactive learning features with AI-assisted gesture recognition. Designed user flows for dictionary, gesture practice, quizzes, and progress tracking and contributed to backend/UI integration.

SightSense — AI, Product Design

Developed a competition project exploring AI and smartphone technology for more accessible early cataract screening. Designed the screening-result and educational experience and collaborated on the product concept and presentation.

SKILLS

Programming: Python, SQL, JavaScript, C, Java | Data: Pandas, Scikit-learn, BigQuery, Power BI, Tableau | Database: MySQL | ETL: Pentaho Data Integration | Development: Next.js, Django, FastAPI | Design: Figma, UI/UX, Prototyping | Tools: GitHub, Google Colab